

Program : Diploma in Printing Technology	
Course Code : 6109C	Course Title: Printing and Package Testing Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3	Periods per semester: 745

Course Objectives:

- To develop tangible knowledge on the various printing & package testing methods.
- To provide functional experience with various printing & package testing methods.
- To be familiarize with printing & package testing tools and equipment.
- To develop precise skill to perform various printing & package testing process.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Packaging testing methods.		Packaging Techniques	4
Printing & Packaging materials properties.		Materials for Printing & Packaging	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To make the use of Printing & Package Testing equipment.	8	Applying
CO2	To perform Printing & Packaging Material Test.	8	Applying
CO3	To perform Mechanical - Tests.	13	Applying
CO4	To perform Tests-Physical, Chemical and Biological.	12	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To make the use of Printing & Package Testing equipment.		
M1.01	Familiarize with various package testing instruments and equipment.	8	Applying
CO2	To perform Printing & Packaging Material Test.		
M2.01	Tensile Testing.	2	Applying
M2.02	Tear Resistance Testing.	2	Applying
M2.03	Puncture Testing.	2	Applying
M2.04	Barrier Property tests.	1	Applying
M2.05	Flexure Testing.	1	Applying
	Lab exam I	2	
CO3	To perform Mechanical - Tests		
M3.01	Drop test.	2	Applying
M3.02	Vibration test.	2	Applying
M3.03	Incline Impact Test.	1	Applying
M3.04	Compression Strength Test.	1	Applying
M3.05	Rolling Test.	2	Applying
M3.06	Drum Test.	2	Applying
M3.07	Abrasion Resistance (Ink Rub Tester) Test.	1	Applying
M3.08	Free Fall Testing.	1	Applying
M3.09	Delaminating or Peel Strength of packaging Paper/board.	1	Applying

C04	To perform Tests-Physical, Chemical and Biological.		
M4.01	Absorption Testing.	2	Applying
M4.02	Accelerated Aging Testing.	2	Applying
M4.03	Atmospheric Conditioning Testing.	2	Applying
M4.04	Infectious Materials Packaging Testing.	2	Applying
M4.05	Water Absorption Testing.	2	Applying
M4.06	Stability Studies Testing.	2	Applying
	Lab exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	Brody Aaron L., “ The Wiley Encyclopedia of Packaging Technology ”, John Wiley & Sons, Inc. New York, 1997.
R1	Hanlon Joseph F, “ Handbook of Package Engineering ”, CRC Press, USA, 1998.
R2	Prakash Shetty, “ Science and Technology of Printing Materials ”, MJP Publishers, Chennai, 2008.
R3	BristonJohn, “ Advances in Plastic Packaging Technology ”, PIRA, UK, 1992.
R4	Chakravarty B, “ A Hand Book for Printing and Packaging Technology ”, Galgotia Publications, 1997.
R5	David Jairus R D, “ Aseptic processing and packaging of food ”, CRC Press, New York, 1996.

Online Resources:

Sl.No	Website Link
1	https://en.wikipedia.org/wiki/Package_testing
2	https://lachenmeier.com/packaging-test
3	https://iip-in.com/research-and-development/testing-of-packaging-materials-and-packages.aspx
4	https://www.thimm.com/products-services/services/packaging-testing